

5	(a) waves overlap / meet / superpose coherence / constant phase difference (<i>not constant λ or frequency</i>) path difference = 0, λ, 2λ or phase difference = 0, 2π, 4π same direction of polarisation/unpolarised	(B1) (B1) (B1) (B1)	max. 3	[3]
	(b) $\lambda = v / f$ $f = 12 \times 10^9 \text{ Hz}$ $\lambda = 3 \times 10^8 / 12 \times 10^9$ (<i>any subject</i>) = 0.025 m	C1 C1 M1 A0		[3]
	(c) maximum at P <u>several</u> minima or maxima between O and P 5 maxima / 6 minima between O and P or 7 maxima / 6 minima including O and P	B1 B1 B1		[3]
	(d) slits made narrower slits put closer together (<i>not just 'make slits smaller'</i>) Allow tilting the slits M1 and explanation of axes of rotation A1	B1 B1		[2]

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